

6-5 Improper Integrals

与无限大有关的积分

let f be integrable on $[a, \infty)$

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx$$

6-6 Numerical Integrals

7 Applications of Integration

7-1 Areas Between Curves

Step

1. 画图形

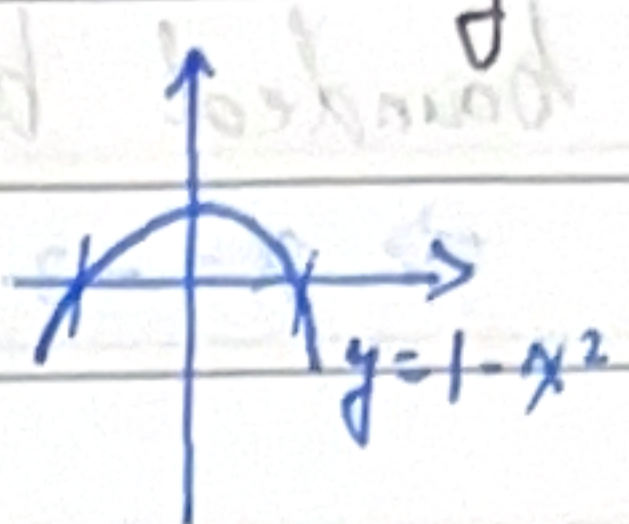
2. 找积分区间

3. 找面积

$$A = \int_a^b [f(x) - g(x)] dx$$

1. Find the area of the region bounded by $y=1-x^2$ and $y=0$.

① $\begin{cases} y=1-x^2 \\ y=0 \end{cases} \Rightarrow x=-1 \text{ or } 1$



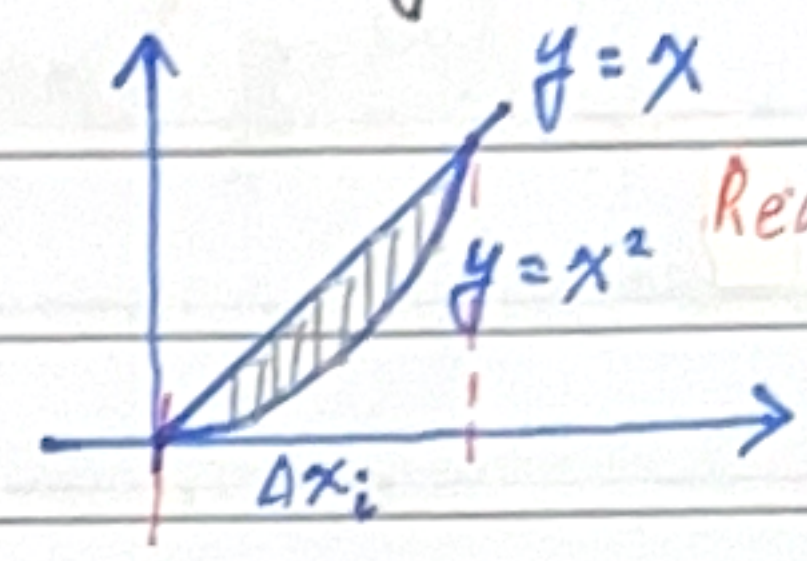
The area is

$$A = \int_{-1}^1 (1-x^2) dx = (x - \frac{1}{3}x^3) \Big|_{-1}^1 = 1 - \frac{1}{3} = \frac{2}{3}$$

2. Find the area of the region bounded by $y=x^2$ and $y=x$

① 方法1
以 x-axis
为切线

$\begin{cases} y=x^2 \\ y=x \end{cases} \Rightarrow x^2-x=0 \Rightarrow x=0 \text{ or } x=1$



Rectangle area = $(f(x) - g(x)) \Delta x_i$

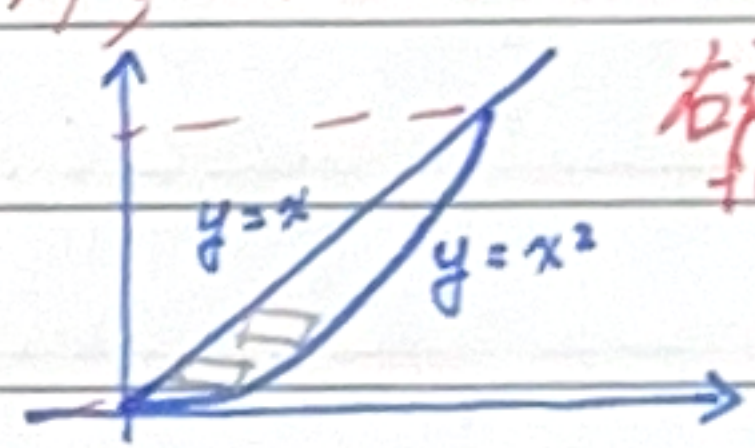
The area is

$$A = \int_0^1 (x-x^2) dx = (\frac{1}{2}x^2 - \frac{1}{3}x^3) \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

上面 - 下面
 $f(x) - g(x)$
长度是 $(f(x)-g(x))$, 宽度是积分变量 (dx)

② 方法2
以 y-axis
为切线

$\begin{cases} y=x^2 \\ y=x \end{cases} \Rightarrow y^2-y=0 \Rightarrow y=0 \text{ or } y=1$



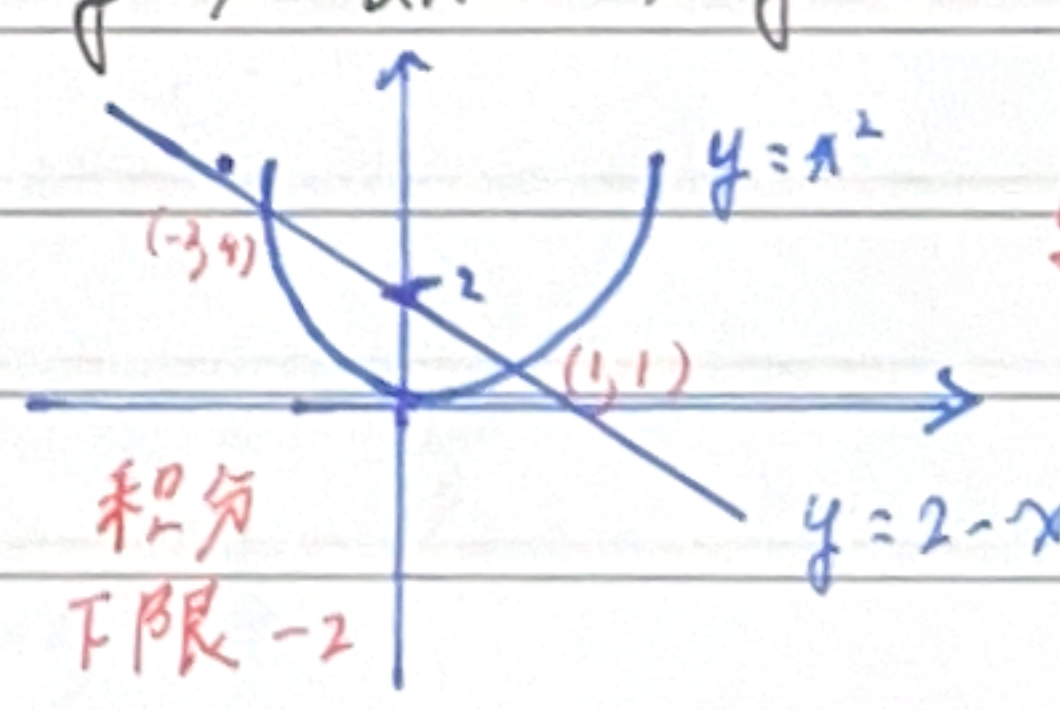
右边 - 左边
 $f(x) - g(x)$

The area is

$$A = \int_0^1 (\sqrt{y} - y) dy = (\frac{2}{3}y^{3/2} - \frac{1}{2}y^2) \Big|_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

3. Find the area of the region bounded by $y=x^2$ and $x+y=2$

① $\begin{cases} y=x^2 \\ y=2-x \end{cases} \Rightarrow x^2=2-x \Rightarrow x^2+x-2=0 \Rightarrow x=-2 \text{ or } x=1$



以 y-axis 为切线
要分段积分
但不是所有情况
可以分段积分

The area is

$$A = \int_{-2}^1 (2-x-x^2) dx = (2x - \frac{1}{2}x^2 - \frac{1}{3}x^3) \Big|_{-2}^1 = 2 - \frac{1}{2} - \frac{1}{3} - (-4 - 2 + \frac{8}{3}) = \frac{9}{2}$$

积分
下限 -2
积分
上限 1

1. Find the area of the region bounded by $y=1-x^2$ and $y=0$

$$\textcircled{2} \begin{cases} y=1-x^2 \Rightarrow x=1 \text{ or } -1 \\ y=0 \end{cases}$$

The area is

$$A = \int_{-1}^1 (1-x^2) dx = \left(x - \frac{1}{3}x^3\right) \Big|_{-1}^1 = 1 - \frac{1}{3} + 1 - \frac{1}{3} = \frac{4}{3}$$

$$\textcircled{2} \begin{cases} y=1-x^2 \Rightarrow x=1 \text{ or } x=-1 \\ y=0 \end{cases}$$

The area is

$$A = \int_{-1}^1 (1-x^2) dx = \left(x - \frac{1}{3}x^3\right) \Big|_{-1}^1 = 1 - \frac{1}{3} + 1 - \frac{1}{3} = \frac{4}{3}$$

2. Find region bounded by $y=x^2$ and $y=x$

$$\textcircled{1} \begin{cases} y=x^2 \Rightarrow x=0 \text{ or } 1 \\ y=x \end{cases}$$

The area is

$$A = \int_0^1 (x-x^2) dx = \left(\frac{1}{2}x^2 - \frac{1}{3}x^3\right) \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

3. Find area bounded by $y=x^2$ and $x+y=2$

$$\textcircled{2} \begin{cases} y=x^2 \Rightarrow x=-2 \text{ or } x=1 \\ y=-x+2 \end{cases}$$

The area is

$$A = \int_{-2}^1 (-x+2-x^2) dx = \left(2x - \frac{1}{2}x^2 - \frac{1}{3}x^3\right) \Big|_{-2}^1 = 2 - \frac{1}{2} - \frac{1}{3} + 4 + 2 - \frac{8}{3} = \frac{9}{2}$$

$$\textcircled{2} \begin{cases} y=x^2 \Rightarrow x=-2 \text{ or } x=1 \\ y=-x+2 \end{cases}$$

The area is

$$A = \int_{-2}^1 (-x+2-x^2) dx = \left(2x - \frac{1}{2}x^2 - \frac{1}{3}x^3\right) \Big|_{-2}^1 = 2 - \frac{1}{2} - \frac{1}{3} + 4 + 2 - \frac{8}{3} = \frac{9}{2}$$

